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Image-Assisted Material Classification for Mobile Dimensioning

Abstract

A method in a computing device includes: capturing sensor data depicting a target object and an adjacent surface; detecting, from the sensor data, a first surface of the target object; identifying, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determining a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and selecting, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of an attribute of the target object.

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Background/Summary

BACKGROUND

[0001] Depth sensors such as time-of-flight (ToF) sensors can be deployed in mobile devices such as handheld computers, and employed to capture point clouds of objects (e.g., boxes or other packages), from which dimensions of the objects can be derived. Point clouds generated by ToF sensors, however, may include artifacts caused by multipath reflections received at the sensor.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0002] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

[0003] FIG. 1 is a diagram of a computing device for determining attributes of an object.

[0004] FIG. 2 is a diagram illustrating a multipath artifact in captured by the mobile computing device of FIG. 1.

[0005] FIG. 3 is a flowchart of a method of image-assisted material classification for mobile dimensioning.

[0006] FIG. 4 is a diagram illustrating a performance of block 305 of the method of FIG. 3.

[0007] FIG. 5 is a diagram illustrating an example performance of block 315 of the method of FIG. 3.

[0008] FIG. 6 is a diagram illustrating a example performance of block 320 of the method of FIG. 3.

[0009] FIG. 7 is a diagram illustrating a example performance of block 325 of the method of FIG. 3.

[0010] FIG. 8 is a diagram illustrating a example performance of block 355 of the method of FIG. 3.

[0011] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0012] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0013] Examples disclosed herein are directed to a method in a computing device, the method comprising: capturing sensor data depicting a target object and an adjacent surface; detecting, from the sensor data, a first surface of the target object; identifying, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determining a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and selecting, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of an attribute of the target object.

[0014] Additional examples disclosed herein are directed to a computing device, comprising: a processor configured to: capture sensor data depicting a target object and an adjacent surface; detect, from the sensor data, a first surface of the target object; identify, from the sensor data, a

material type of the adjacent surface; for a sample point on the first surface of the target object, determine a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and select, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of an attribute of the target object.

[0015] Further examples disclosed herein are directed to a non-transitory computer-readable medium storing computer-readable instructions executable by a processor of a computing device to: capture sensor data depicting a target object and an adjacent surface; detect, from the sensor data, a first surface of the target object; identify, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determine a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and select, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of an attribute of the target object.

[0016] FIG. 1 illustrates a computing device **100** configured to capture sensor data depicting a target object **104** within a field of view (FOV) of a sensor of the device **100**. The computing device **100**, in the illustrated example, is a mobile computing device such as a tablet computer, smartphone, or the like. The computing device **100** can be manipulated by an operator thereof to place the target object **104** within the FOV of the sensor, in order to capture sensor data for subsequent processing as described below. In other examples, the computing device **100** can be implemented as a fixed computing device, e.g., mounted adjacent to an area in which target objects **104** are placed and/or transported (e.g., a staging area, a conveyor belt, a storage container, or the like).

[0017] The target object **104**, in this example, is a parcel (e.g., a cardboard box or other substantially cuboid object), although a wide variety of other target objects can also be processed as set out below, including non-cuboid objects. The target object **104** is shown resting on a support surface **108** (e.g., a floor, table, conveyor, or the like), with an adjacent surface **112** nearby, such as a wall, a door, another parcel or stack of parcels, or the like.

[0018] The sensor data captured by the computing device **100** includes a point cloud and/or depth image. The point cloud includes a plurality of depth measurements (also referred to as points) defining three-dimensional positions of corresponding points on the target object **104**. The sensor data captured by the computing device **100** also includes a two-dimensional (2D) image depicting the target object **104**. The 2D image can include a two-dimensional array of pixels, each pixel containing a color and/or brightness value. For instance, the image can be a color image in which each pixel in the array contains a plurality of color component values (e.g., values for red, green and blue levels, or for any other suitable color model). From the captured sensor data, the device **100** (or in some examples, another computing device such as a server, configured to obtain the sensor data from the device **100**) is configured to determine dimensions of the target object **104**, such as a width “W”, a depth “D”, and a height “H” of the target object **104**.

[0019] When the target object **104** is non-cuboid, the dimensions of the target object **104** need not align with physical edges of the object **104**. For example, dimensions of an irregularly shaped object can be the width, depth, and height of a cuboid space encompassing the object **104**, or the dimensions can include various other measurements of the object **104**. The computing device **100** can further be configured to determine other attributes of the object **104** in addition to or instead of the dimensions noted above. For example, the computing device **100** can be configured to classify the object **104** into various types based on captured sensor data, to detect a location of the object **104**, or the like.

[0020] The target object **104** is, in the examples discussed below, a substantially rectangular prism. As shown in FIG. 1, the height H of the object **104** is a dimension substantially perpendicular to a support surface (e.g., a floor) **108** on which the object **104** rests. The width W and depth D of the object **104**, in this example, are substantially orthogonal to one another and to the height H. The

dimensions determined from the captured data can be employed in a wide variety of downstream processes, such as optimizing loading arrangements for storage containers, pricing for transportation services based on parcel size, and the like.

[0021] Certain internal components of the device **100** are also shown in FIG. **1**. For example, the device **100** includes a processor **116** (e.g., a central processing unit (CPU), graphics processing unit (GPU), and/or other suitable control circuitry, microcontroller, or the like). The processor **116** is interconnected with a non-transitory computer readable storage medium, such as a memory **120**. The memory **120** includes a combination of volatile memory (e.g. Random Access Memory or RAM) and non-volatile memory (e.g. read only memory or ROM, Electrically Erasable Programmable Read Only Memory or EEPROM, flash memory). The memory **120** can store computer-readable instructions, execution of which by the processor **116** configures the processor **116** to perform various functions in conjunction with certain other components of the device **100**. The device **100** can also include a communications interface **124** enabling the device **100** to exchange data with other computing devices, e.g. via various networks, short-range communications links, and the like.

[0022] The device **100** can also include one or more input and output devices, such as a display **128**, e.g., with an integrated touch screen. In other examples, the input/output devices can include any suitable combination of microphones, speakers, keypads, data capture triggers, or the like.

[0023] The device **100** further includes a depth sensor **132**, controllable by the processor **116** to capture point clouds. The device **100** also includes an image sensor, also referred to as a camera **136**, configured to capture two-dimensional images, such as the color images mentioned above. In some examples, the depth sensor **132** and the camera **136** can be implemented by a single sensor device configured to capture both depth measurements for generating point clouds, and color and/or intensity measurements for generating 2D images.

[0024] The depth sensor **132** can include a time-of-flight (ToF) sensor, e.g., mounted on a housing of the device **100**, for example on a back of the housing (opposite the display **128**, which is visible in FIG. **1**) and having an optical axis that is substantially perpendicular to the display **128**. A ToF sensor can include, for example, an emitter (e.g., a laser emitter) configured to illuminate a scene, and an image sensor configured to capture light from the emitter as reflected by the scene. The ToF sensor can further include a controller configured to determine a depth measurement for each captured reflection, according to the time difference between illumination pulses and reflections. Each depth measurement indicates a distance between the depth sensor **132** itself and the point in space where the reflection originated. Each depth measurement represents a point in a resulting point cloud. The depth sensor **132** and/or the processor **116** can be configured to convert the depth measurements into points in a three-dimensional coordinate system **140**. Although the coordinate system **140** is shown with an origin on the support surface **108**, a wide variety of other coordinate systems can also be used, e.g., with an origin at the depth sensor **132**.

[0025] As will be apparent to those skilled in the art, the depth sensor **132** can also be configured to generate 2D images, e.g., by capturing both the reflections from emitted light and ambient light, and generating a two-dimensional array of pixels containing intensity values. For illustrative purposes, however, 2D images processed in the discussion below are captured by the camera **136**, e.g., simultaneously with the capture of point clouds by the depth sensor **132**. The camera **136** may, in some examples, produce a 2D image with a greater resolution than the depth sensor **132** (e.g., with a greater number of pixels representing a given portion of the scene). The points of the point cloud can be mapped to corresponding pixels of the 2D image according to a transform defined by calibration data for the sensors **132** and **136** (e.g., sensor extrinsic and intrinsic matrices).

[0026] The memory **120** stores computer readable instructions for execution by the processor **116**. In particular, the memory **120** stores a dimensioning application **144** which, when executed by the processor **116**, configures the processor **116** to process one or more point clouds (e.g., one or more successive frames of depth measurements, converted to point clouds representing the target object

104 and surroundings at successive points in time) to detect the target object **104** and determine dimensions (e.g., the width, depth, and height shown in FIG. 1) and/or other attributes of the target object **104**, such as an object type or class, an object location, or the like. For example, the dimensioning process implemented by the application **144** can include identifying a first surface of the object **104**, such as an upper surface **148**, as well as the support surface **108**, in the point cloud. The height H of the target object **104** can be determined as the distance between the upper surface **148** and the support surface **108** (e.g., the perpendicular distance between the surfaces **148** and **108**). The width W and the depth D can be determined as the dimensions of the upper surface **148**. [0027] Under some conditions, the point cloud captured by the depth sensor **132** can contain artifacts that impede the determination of accurate dimensions of the target object **104**. For example, multipath artifacts can cause certain portions of the target object **104** to appear further from the sensor than those portions are in reality, which can lead to a point cloud containing a distorted representation of the target object **104**.

[0028] An example multipath scenario is illustrated in FIG. 1, in which light emitted by the sensor **132** (shown in longer dashes) reflects from a point **150** on the upper surface **148** back towards the sensor **132**, as well as from the point **150** towards a point **152** on the wall **112**. At least a portion of the light reflected from the point **150** and impacting the point **152** may reflect (reflections are shown with shorter dashes) toward the sensor **132**, such that the reflections received at the sensor **132** corresponding to the point **150** include both a reflection directly from the point **150**, and a multipath reflection from the point **152**. The reflections detected by the sensor **132** corresponding to the point **150** may therefore inflate the perceived distance from the sensor **132** to the point **150**. Certain surfaces of the target object **104** may therefore appear curved or otherwise distorted in the point cloud.

[0029] The device **100** is therefore configured to implement additional functionality to detect when a point cloud captured via the sensor **132** (e.g., one frame of point cloud data) is likely to contain multipath artifacts sufficient to distort the appearance of the target object **104**. The device **100** can further select handling actions for captured point clouds based on the above detection. For example, when the device **100** detects such artifacts, the device **100** can suppress dimensioning of the target object **104** based on the captured point cloud, e.g., to instead capture a further point cloud, generate a notification instructing an operator of the device **100** to move the target object **104**, or the like. The functionality discussed herein can also be implemented via execution of the application **144** by the processor **116**. In other examples, some or all of the functionality described herein can be performed via dedicated hardware (e.g., an application-specific integrated circuit or ASIC, or the like), or by a distinct computing device such as a server in communication with the device **100**.

[0030] As discussed below, the additional functionality implemented by the device **100** includes identifying, from a two-dimensional image captured substantially simultaneously with the point cloud (e.g., via the camera **136**), a material type of the wall **112** or any other adjacent surfaces to the target object **104**. The device **100** can maintain, e.g., in the memory **120**, a repository **156** of reflectivity coefficients for one or more material types. The device **100** can therefore, based on the identified material type of a surface adjacent to the target object **104** (such as the wall **112**), determine a reflection intensity for reflections from the adjacent surface that may contribute to multipath artifacts in the point cloud. For example, the repository **156** can store, for each of a set of material types, a specular reflectivity coefficient, a specular exponent (also referred to as a shininess parameter), and a diffuse reflectivity coefficient.

[0031] Referring to FIG. 2, an overhead view of the device **100**, the target object **104**, and the wall **112** is shown. Following emission of a pulse of illumination by the sensor **132**, a single pixel of the sensor **132** may receive two distinct reflections. A first reflection **200-1** results directly from emitted light **204-1** impacting the point **150**. A second reflection **200-2**, however, results from a portion of the emitted light **204-1** generating a reflection **204-2** from the point **150** towards the point **152** on the wall **112**, and then reflecting to the sensor **132**. The sensor **132** can integrate the

reflections **200** to generate an aggregated depth measurement corresponding to the point **150**. Due to the variable nature of multipath reflections, however, it may be difficult to accurately determine the position of the point **150** in three-dimensional space. For example, the sensor may overestimate the distance between the sensor and the point **150**. The resulting point cloud, for instance, may depict an upper surface **148'** that is distorted relative to the true shape of the upper surface **148** (the object **104** is shown in dashed lines below the surface **148'** for comparison). The surface **148'**, in this exaggerated example, has a curved profile and is larger in one dimension than the true surface **148**. Multipath artifacts in captured point clouds may therefore lead to inaccurate dimensions for the object **104**.

[0032] Turning to FIG. **3**, a method **300** of image-assisted material classification for mobile dimensioning is illustrated. The method **300** is described below in conjunction with its performance by the device **100**, e.g., to dimension the target object **104**. It will be understood from the discussion below that the method **300** can also be performed by a wide variety of other computing devices including or connected with sensor assemblies functionally similar to the sensor assembly **132**.

[0033] At block **305**, the device **100** is configured, e.g., via control of the depth sensor **132** and the camera **136** by the processor **116**, to capture a point cloud depicting at least a portion of the object **104**, and a two-dimensional image depicting at least a portion of the object **104**. The device **100** can, for example, be positioned relative to the object **104** as shown in FIG. **1**, to capture a point cloud and image depicting the upper surface **148** and one or more other surfaces of the object **104**. The point cloud and the 2D image are captured substantially simultaneously, e.g., by triggering the depth sensor **132** and the camera **136** at substantially the same time. In some examples, in which the depth sensor **132** and the camera **136** are implemented as a single sensor, simultaneous capture of the 2D image and the point cloud can be implemented by triggering such a combined sensor.

[0034] FIG. **4** illustrates an example point cloud **400** and an example image **404** captured at block **305**. The point cloud **400** defines a plurality of points each having three-dimensional positions, e.g., specified in the coordinate system **140**. Although outlines are shown for the object **104**, the support surface **108**, and the wall **112** for clarity, it will be understood that the point cloud **400** in practice does not include such outlines, object boundaries, or the like. The device **100** is instead configured, as discussed below, to distinguish objects in the point cloud from one another based on plane fitting operations and/or other suitable segmentation mechanisms. It will also be understood that the density of points in the point cloud **400** need not be consistent throughout the point cloud, as shown in FIG. **4**. For example, certain regions of the point cloud **400** may have lower point densities than others.

[0035] The image **404** includes a plurality of pixels each having color and/or intensity values. The image **404** may include regions of similar color and/or intensity corresponding to distinct objects in the scene, e.g., as a result of the different materials constituting those objects. Such regions are indicated in FIG. **4** by different fill patterns, e.g., including a region **408** corresponding to the wall **112**, a region **412** corresponding to the object **104**, and a region **416** corresponding to the support surface **108**. It will be understood that in reality a given surface (such as the wall **112**) may not appear as consistently as shown in FIG. **4**, due to shadows and other ambient lighting effects, variations in surface texture, and the like.

[0036] Returning to FIG. **3**, at block **310** the device **100** is configured to detect at least one surface of the target object **104** in the point cloud **400**. The device **100** can also detect other surfaces in the point cloud **400**, such as the support surface **108** and the wall **112**. In the present example, the device **100** is configured to detect at least the upper surface **148** of the target object **104**, as well as the support surface **108** and the wall **112**. The device **100** can also detect other surfaces of the object **104**, such as the side and front surfaces visible in FIGS. **1** and **4**.

[0037] To detect surfaces in the point cloud **400**, the device **100** can be configured, e.g., via execution of the application **144**, to perform a plane-fitting algorithm, such as random sample

consensus (RANSAC), or the like. The support surface **108** can be distinguished from other surfaces during such detection by, for example, selecting the detected surface with the lowest height (e.g., the lowest Z value in the coordinate system **140**). The upper surface **148** can be distinguished from other surfaces during detection by, for example, selecting a surface substantially parallel to the support surface **108** and/or substantially centered in the point cloud **400**. The wall **112** can be distinguished from other surfaces in the point cloud **400** by its orientation (e.g., being substantially orthogonal to the support surface **108** and the upper surface **148**). In some examples, the detection of the object **104** and/or other surfaces in the point cloud **400** can include processing the 2D image from block **305** via a machine-learning based object classifier, e.g., to obtain bounding boxes, masks, or the like corresponding to one or more of the object **104**, the support surface **108**, and the wall **112**. Such bounding boxes can be used as inputs to plane fitting operations or the like.

[0038] At block **315**, the device **100** is configured to identify, from the 2D image, a material type of at least one surface adjacent to the target object **104**. For example, the application **144** can include an image classification model, or such a model can be implemented as a distinct application executable by the processor **116** and called via the application **144**, for example. When executed by the processor **116**, the segmentation model configures the processor **116** to select one of a set of predetermined material types for each of a plurality of regions in the image **404**. For example, the processor **116** can assign a material type to each pixel of the image **404**, resulting in regions of neighboring pixels with common material types. The model, such as a Region-based Convolutional Neural Network (R-CNN), a Fully Convolutional Network (FCN), or the like, can be trained with a set of images in which each pixel and/or other suitable region is labelled with material types.

[0039] The material types selected by the segmentation model can be determined during deployment of the application **144**, for example, and can include materials likely to appear in the images captured at block **305**. Example materials can include, for example, cardboard (e.g., if the object **104** is generally a cardboard box or other package), various support surface materials such as polished concrete, carpet or the like, and various wall surface materials, such as a matte white paint, or the like.

[0040] FIG. **5** illustrates an example result of the performance of block **315** for the image **404**. In particular, each pixel of the image **404** can be labelled with one of the predetermined material identifiers mentioned above. In this example, a label **500** corresponds to a material identifier “white paint”, a label **504** corresponds to a material identifier “cardboard”, and a label **508** corresponds to a material identifier “blue carpet”. The labels **500**, **504**, and **508** need not be text strings in other examples, but can instead include numerical identifiers. Although the exact appearance of each pixel may vary, e.g., such that two pixels corresponding to the wall **112** may have difference color and/or intensity values, the segmentation model is configured to identify the appropriate material type (e.g., white paint, in this case) despite such differences. The output of block **315** can be a labelled version of the image **404**, e.g., in which each pixel has a material type associated therewith.

[0041] Returning to FIG. **3**, following the detection of object surfaces from the point cloud at block **310**, and the identification of material types at block **315**, beginning at block **320** the device **100** is configured to combine the detected surfaces and the identified material types to assess whether the point cloud captured at block **305** is likely to be affected by multipath artifacts to a degree sufficient to impede dimensioning of the target object **104**.

[0042] At block **320**, the device **100** is configured to select at least one sample point on a surface of the object **104**, from the point cloud **400**. In this example, the device **100** is configured to select one or more sample points on the surface **148** of the target object **104**. In other examples, the device **100** can also select sample point(s) on other surfaces of the object **104** that are depicted in the point cloud **400**.

[0043] Turning to FIG. **6**, the device **100** is configured to select a plurality of sample points **600**,

including a sample point **600-1** discussed in further detail below, on the upper surface **148** of the object **104**. The sample points **600** can be selected to form a grid or other predetermined pattern, or can be selected randomly. The number of sample points **600** selected at block **320** can be predefined, e.g., as a configuration setting of the application **144**. In other examples, the number of sample points **600** can be determined based on a sample point density setting, such that larger target objects **104** lead to larger numbers of selected sample points.

[0044] Referring again to FIG. 3, at blocks **325** to **345**, the device **100** is configured to assess the sample points from block **320** and determine whether the sample points are likely to result in multipath artifacts in the point cloud **400**. The assessment represented by blocks **325** to **345** is repeated for each sample point selected at block **320**.

[0045] At block **325**, the device **100** is configured to determine whether a specular reflection is likely to have originated at a surface other than the surface carrying the sample point, impacted the sample point, and contributed to light received at the depth sensor **132** from the sample point **600**. As will be apparent, such a reflection may distort the position of the sample point **600** observed by the sensor **132**.

[0046] To perform block **325**, the device **100** can be configured to determine whether a trajectory exists from the sensor **132**, to a point on a surface distinct from the object **104** (e.g., the wall **112**), to the sample point under consideration. Referring to FIG. 7, a side view of the wall **112** and a portion of the object **104** is shown, with the sample point **600-1** indicated. The device **100** can be configured to generate a plurality of rays **700** originating at the sample point **600-1**. Four rays **700** are illustrated in FIG. 7, although it will be understood that a greater number of rays **700** can also be generated. For example, the device **100** can generate a plurality of rays distributed over a conical region **704** (illustrated in perspective view in FIG. 6) originating at the sample point **600-1**.

[0047] The device **100** can generate a hemispherical region centered on the sample point **600-1**, but as will be apparent, rays originating at the sample point **600-1** and travelling towards the sensor **132**, rather than towards the wall **112** or any other surfaces, are unlikely to yield multipath reflections. Therefore, the computational load associated with reflection assessments for each ray can be reduced with little or no effect on the accuracy of the assessment by using a conical region, oriented away from the sensor **132** (e.g., oriented as shown in FIG. 6).

[0048] For each ray generated for the sample point **600-1**, the device **100** is configured to determine whether the ray **700** impacts another surface represented in the point cloud. When that determination is negative, the ray **700** can be discarded. When the determination is affirmative, as in the case of the four example rays **700** shown in FIG. 7, which all impact the wall **112**, the processor **116** is configured to determine whether a ray between the sensor **132** and the point of impact is likely to generate a reflection that coincides with the ray **700** under consideration. In other words, the processor **116** determines whether an angle between a normal of the wall **112** and a ray from the sensor **132** to the point of impact is substantially equal to an angle between the normal and the ray **700** (e.g., within a predetermined threshold). Alternatively, the processor **116** can determine a reflection vector for the ray **700** at the impacted point, and then determine whether the reflection vector intersects the sensor **132**.

[0049] As seen in FIG. 7, particularly in the detail view **708**, the processor **116** can determine an angle between the ray **700-1** and a normal vector **712** of the wall **112** (at the point **152**), and can also determine a reflection vector **716** corresponding to the ray **700-1**. The reflection vector **716**, which has an equal angle to the normal **712** as the ray **700-1**, intersects the sensor **132**. The ray **700-1** therefore is a potential source of a specular reflection from the point **152** of the wall **112**. The other example rays **700** shown in FIG. 7, however, are not potential sources of specular reflections at the wall **112**.

[0050] For any ray(s) that are potential sources of specular reflections, at block **330** the device **100** is configured to determine a specular reflection intensity for those rays. The determination of a specular reflection intensity can be made according to the following specular reflection model:

$$[00001] I_{\text{spec}} = K_s I_p (\cos \theta)^n \quad \text{Equation(1)}$$

[0051] In equation (1) above, I_{spec} is the intensity of a specular reflection, K_s is a specular reflectivity coefficient of the wall **112** (or any other surface impacted by a ray **700**), I_p is the intensity of the light source (that is, the emitter of the sensor **132**). Further, the angle θ (theta) is the angle between the normal **712** and the ray **700**, and the constant n is the specular exponent, also referred to as a shininess parameter. The intensity of the light source can be retrieved from manufacturer data for the sensor **132** stored in the memory **120**. The angle θ is derived from the point cloud **400** and the relevant one of the rays **700**. The specular reflectivity coefficient and the specular exponent are retrieved from the repository **156**, based on a material type identified at block **315**. In particular, the processor **116** is configured to determine which pixel of the image **404** corresponds to the point **152** (in this example) impacted by the ray **700-1**, based on the transform defined by calibration data for the sensors **132** and **136** as mentioned earlier. The processor **116** is then configured to determine which material type was identified for the corresponding pixel of the 2D image, and to retrieve the specular reflectivity coefficient and specular exponent for that material type from the repository **156**. As will be apparent from equation (1), the intensity of a specular reflection, for a given material, decreases for shallower impact angles.

[0052] When the determination at block **325** is negative, the processor **116** determines a diffuse reflection intensity for the ray **700** under consideration. Diffuse reflection intensity can be made according to the following diffuse reflection model:

$$[00002] I_{\text{diff}} = K_d I_p \cos \theta \quad \text{Equation(2)}$$

[0053] In equation (2) above, I_{diff} is the intensity of a diffuse reflection, K_d is a diffuse reflectivity coefficient of the wall **112** (or any other surface impacted by a ray **700**). As in equation (1), I_p is the intensity of the light source and θ (theta) is the angle between the normal **712** and the ray **700**. As will now be apparent, the diffuse reflectivity coefficient can be retrieved for the corresponding material type from the repository **156**.

[0054] Following block **330** (if the determination at block **325** was affirmative) or block **335** (if the determination at block **325** was negative), the device **100** proceeds to block **340**. At block **340**, the device **100** is configured to determine whether the intensity from block **330** or **335** exceeds an intensity threshold. The threshold can be a predetermined intensity, e.g., in candela or any other suitable unit of measurement. When the determination at block **340** is negative for the current ray, the processor **116** is configured to continue processing the remaining rays for the current sample point **600**, as well as any remaining sample points **600**.

[0055] When the determination at block **340** is affirmative, at block **345** the device **100** is configured to increment a multipath artifact score corresponding to the point cloud. In other words, the multipath artifact score is incremented for each ray originating from each sample point that is likely to produce a sufficiently intense multipath reflection. The device **100** then continues processing the remaining rays **700** and sample points **600**, until each ray **700** of each sample point **600** has been processed. In other examples, the determination at block **340** can be made once all of the rays for a given sample point are assessed, such that the threshold applied at block **340** represents an aggregated reflection intensity for a given sample point. The multipath score can therefore be incremented once for each sample point **600**, based on the collected reflection intensities determined for that sample point **600**.

[0056] Following block **345**, or following a negative determination at block **340**, the device **100** is configured, at block **348**, to determine whether any sample points **600** remain to be processed. When the determination at block **348** is affirmative, the device **100** returns to block **325** to process the next sample point. When the determination at block **348** is negative, indicating that all sample points **600** have been processed as described above, the device **100** proceeds to block **350**.

[0057] At block **350**, the device **100** is configured to determine whether the multipath artifact score accumulated through multiple performances of blocks **325** to **345** exceeds a predetermined

threshold. The processor **116** is configured to select a handling action for the point cloud **400** based on the determination at block **350**. In particular, the processor **116** is configured to select between dimensioning the target object **104** using the point cloud **400**, and suppressing dimensioning of the target object **104** until a further point cloud is captured (e.g., in the next frame of data from the depth sensor **132** and the camera **136**).

[0058] When the determination at block **350** is affirmative, the potential contributions of multipath reflections from adjacent surfaces such as the wall **112** are sufficient that the accuracy of the point cloud **400** may be negatively affected, and the device **100** can suppress dimensioning of the object **104** using the point cloud **400**. Instead, the device **100** may generate a notification, e.g., on the display **128**, an audible tone, as a message to another computing device, or the like, indicating that dimensioning has been suppressed. FIG. **8** illustrates an example notification **800** generated at block **355**. In other examples, the notification can include an instruction to an operator of the device **100** to reposition the object **104** away from nearby surfaces.

[0059] At block **360**, in response to a negative determination at block **350**, the device **100** can proceed to determine one or more attributes of the object **104**, such as dimensions of the object **104**, for presentation on the display **128**, transmission to another computing device, processing via a further application executed by the device **100**, or the like.

[0060] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.

[0061] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0062] Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0063] Certain expressions may be employed herein to list combinations of elements. Examples of such expressions include: “at least one of A, B, and C”; “one or more of A, B, and C”; “at least one of A, B, or C”; “one or more of A, B, or C”. Unless expressly indicated otherwise, the above expressions encompass any combination of A and/or B and/or C.

[0064] It will be appreciated that some embodiments may be comprised of one or more specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used.

[0065] Moreover, an embodiment can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g., comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

[0066] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A method in a computing device, the method comprising: capturing sensor data depicting a target object and an adjacent surface; detecting, from the sensor data, a first surface of the target object; identifying, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determining a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and selecting, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of the attribute of the target object.
2. The method of claim 1, wherein determining the material type of the adjacent surface includes: executing a segmentation model to determine, for each pixel of the two-dimensional image, one of a set of predetermined material types; and storing the determined material type for each pixel.
3. The method of claim 2, further comprising: storing, for each of the predetermined material types, a reflectivity coefficient; and determining the reflection intensity based on the stored reflectivity coefficient for the material type of the adjacent surface.
4. The method of claim 3, wherein storing the reflectivity coefficient includes storing a specular reflectivity coefficient and a diffuse reflectivity coefficient; and selecting, based on an angle of incidence of a ray between the sample point and the adjacent surface, between the specular

reflectivity coefficient and the diffuse reflectivity coefficient; and determining the reflection intensity based on the selected one of the specular reflectivity coefficient and the diffuse reflectivity coefficient.

5. The method of claim 1, further comprising: in response to selecting determining the attribute of the target object: determining dimensions of the target object from the point cloud based on the first surface; and presenting the dimensions on a display of the computing device.

6. The method of claim 1, further comprising: in response to selecting suppressing the determination of the attribute of the target object, generating a notification indicating that the point cloud likely contains multipath artifacts.

7. The method of claim 1, further comprising: determining reflection intensities for each of a plurality of sample points on the first surface; incrementing a multipath score for each reflection intensity that exceeds a first threshold; and selecting the handling action by comparing the multipath score to a second threshold.

8. The method of claim 7, wherein selecting the handling action includes selecting suppressing dimensioning of the target object when the multipath score exceeds the second threshold.

9. The method of claim 1, wherein the sensor data includes a point cloud and a two-dimensional image; and wherein the first surface of the target object is detected from the point cloud, and the material type of the adjacent surface is determined from the two-dimensional image.

10. A computing device, comprising: a processor configured to: capture sensor data depicting a target object and an adjacent surface; detect, from the sensor data, a first surface of the target object; identify, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determine a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and select, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of the attribute of the target object.

11. The computing device of claim 10, wherein the processor is configured to determine the material type of the adjacent surface by: executing a segmentation model to determine, for each pixel of the two-dimensional image, one of a set of predetermined material types; and storing the determined material type for each pixel.

12. The computing device of claim 11, wherein the processor is further configured to: store for each of the predetermined material types, a reflectivity coefficient; and determine the reflection intensity based on the stored reflectivity coefficient for the material type of the adjacent surface.

13. The computing device of claim 12, wherein the processor is further configured to: store, for each of the predetermined material types, a specular reflectivity coefficient and a diffuse reflectivity coefficient; and select, based on an angle of incidence of a ray between the sample point and the adjacent surface, between the specular reflectivity coefficient and the diffuse reflectivity coefficient; and determine the reflection intensity based on the selected one of the specular reflectivity coefficient and the diffuse reflectivity coefficient.

14. The computing device of claim 10, wherein the processor is further configured to: in response to selecting suppressing the determination of the attribute of the target object, generate a notification indicating that the point cloud likely contains multipath artifacts.

15. The computing device of claim 10, wherein the processor is further configured to: determine reflection intensities for each of a plurality of sample points on the first surface; increment a multipath score for each reflection intensity that exceeds a first threshold; and select the handling action by comparing the multipath score to a second threshold.

16. The computing device of claim 15, wherein the processor is configured to select the handling action by selecting suppressing the determination of the attribute of the target object when the multipath score exceeds the second threshold.

17. The computing device of claim 10, wherein the sensor data includes a point cloud and a two-dimensional image; and wherein the first surface of the target object is detected from the point

cloud, and the material type of the adjacent surface is determined from the two-dimensional image.

18. A non-transitory computer-readable medium storing computer-readable instructions executable by a processor of a computing device to: capture sensor data depicting a target object and an adjacent surface; detect, from the sensor data, a first surface of the target object; identify, from the sensor data, a material type of the adjacent surface; for a sample point on the first surface of the target object, determine a reflection intensity from the adjacent surface based on the material type determined from the sensor data; and select, based on the reflection intensity, a handling action from (i) determining an attribute of the target object and (ii) suppressing the determination of the attribute of the target object.
